

# Mohamed Adem BEN MOUSSA

Database & Platform Engineer | PostgreSQL • Elasticsearch • Linux • ETL • AML/KYC

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Tunis, Tunisia (Open to relocate)



## PROFESSIONAL SUMMARY

Database & Platform Engineer with 5+ years of experience in fintech, ensuring production stability and data integrity for AML/KYC platforms used by 250+ financial institutions across 45+ countries. Specialized in PostgreSQL, ETL pipelines, distributed systems and production infrastructure, with a strong focus on high availability and performance optimization. Resolved 2,000+ production incidents at 93% SLA compliance across 32 environments. Ranked 1st in class (2022–2023).

## PROFESSIONAL EXPERIENCE

### Database & Systems Engineer

Jan 2021 – Present

Vneuron, Chargaia 2, Tunisia

- Lead production stability and data integrity across 32 high-tier financial institutions (portfolio of 117 clients, 250+ globally), ensuring high availability and SLA compliance.
- Developed PL/SQL batch processing scripts to safely purge 96M+ orphaned records without table locks, reducing execution from hours to under 13 minutes.
- Resolved 2,000+ production incidents at 93% SLA compliance, implementing root-cause fixes and improving system reliability.
- Optimized PL/SQL queries and indexing strategies on multi-million row tables, reducing critical operation times from minutes to seconds.
- Managed multi-tenant PostgreSQL deployments, automated backups, and tuned server configurations for cloud and on-prem environments.
- Maintained Elasticsearch clusters for real-time KYC sanction screening under continuous high-volume ingestion.
- Designed and maintained AML/KYC risk indicator rules, Camunda BPMN workflows, and FormIO forms per client requirements.
- Built and maintained Talend ETL pipelines and automated monthly Jaspersoft Studio reports for 117 clients.
- Administered Apache Tomcat and Nginx, including SSL, log monitoring, and system hardening.
- Implemented automated deployment and upgrade processes for Apache Tomcat on RHEL, improving consistency and reducing manual intervention.
- Mentored junior engineers, documented internal procedures in Confluence, and maintained client guides and personal knowledge bases.
- Leveraged monitoring, alerting, performance optimization, and automation practices to ensure system reliability and uptime.

Technologies: PostgreSQL, PL/SQL, Elasticsearch, Talend ETL, Camunda BPMN, FormIO, Reis™ Rule Engine, Tomcat, Nginx, Jaspersoft Studio, JIRA, Confluence, Linux, Bash, PowerShell, Postman, pgAdmin, DBeaver, pgtune, SQLBackupAndFTP

### Software Integration Consultant

Jul 2020 – Jan 2021

Vneuron, Chargaia 2, Tunisia

- Deployed and configured AML/KYC platforms for new clients, including PostgreSQL, Elasticsearch, Apache Tomcat, and Nginx environments.
- Built and executed Talend ETL pipelines to migrate and integrate client data into production systems.
- Configured AML risk indicators, Camunda BPMN workflows, and FormIO forms based on client requirements.
- Developed compliance dashboards and reporting solutions using Jaspersoft Studio.
- Automated PostgreSQL backup processes using Bash and PowerShell, improving reliability and reducing manual intervention.
- Supported testing, validation, and client acceptance phases, ensuring smooth production rollouts.
- Analyzed client requirements and contributed to technical specifications and solution design.
- Produced technical documentation for deployed solutions and configurations.

Technologies: PostgreSQL, Talend ETL, Elasticsearch, Tomcat, Nginx, FormIO, Camunda BPMN, Jaspersoft Studio, JavaScript, Postman, NSSM, Linux, Bash, PowerShell, Windows

## EDUCATION

### National Engineering Degree in Computer Science

Oct 2020 – Sep 2024

SESAME University, Ariana, Tunisia

- GPA: 3.5/4.0 · Ranked 1st in class, 2022-2023 · Final year: 16.86/20 (4.0/4.0)
- Official equivalence recognized by Ministry of Higher Education (Decision N°1555, Feb 2025)
- Graduation Project: Securing an AML Suite with an Advanced IDS (Suricata + Wazuh)

### Bachelor's Degree in Computer Science

Sep 2017 – Jul 2020

ISTIC University, Borj Cédria, Tunisia

- Graduation Project: Remote Control Light Switch Using IoT Technology (Arduino, Raspberry Pi)

## TECHNICAL SKILLS

|                          |                                                                                                                                   |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Databases:               | PostgreSQL (PL/SQL, indexing, query optimization, batch processing, multi-tenant), MySQL, Oracle, SQL Server, MariaDB             |
| Search / Analytics:      | Elasticsearch (installation, service management, JVM heap tuning, REST API), Jaspersoft Studio, Talend ETL                        |
| Infrastructure:          | Linux (RHEL, Ubuntu, CentOS), Windows Server, Apache Tomcat, Nginx, Systemd, High Availability, Monitoring, Backup/Recovery       |
| Workflows & Rules:       | Camunda BPMN, FormIO, Reis™ Rule Engine, AML/KYC Workflows, Risk Indicator Rules                                                  |
| Practices:               | Incident Management, Production Support, Automation, Performance Optimization, Deployment, System Reliability, Logging/Monitoring |
| Security:                | Suricata (IDS), Wazuh (SIEM), Permissions Hardening, Vulnerability Analysis                                                       |
| Identity & Access:       | Active Directory, LDAP, GLPI, Role-based Access Control                                                                           |
| Scripting / Development: | Bash, PowerShell, JavaScript, PHP, Java                                                                                           |
| Tools / Platforms:       | JIRA, Confluence, Notion, Postman, pgAdmin, DBeaver, pgTune, SQLBackupAndFTP                                                      |

## CERTIFICATIONS

Information Technology Specialist in Databases – Certiport (Jun 2022, License: [w96Fd-FaGp](#))

## LANGUAGES

Arabic: Native, English: Professional Working, French: Professional Working

## PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Automated Apache Tomcat Upgrade &amp; Deployment Script (RHEL)</b> <ul style="list-style-type: none"><li>Developed a Bash automation script to install and upgrade Apache Tomcat (9.x) across production servers.</li><li>Automated configuration migration, permission management, and systemd service updates.</li><li>Reduced manual deployment effort, improved consistency, and minimized configuration drift across environments.</li></ul>                                                                                     | 2025 |
| <b>Securing an AML/KYC Platform with IDS &amp; SIEM</b> – Vneuron, Charguia 2, Tunisia <ul style="list-style-type: none"><li>Deployed Suricata (IDS) and Wazuh (SIEM) to secure a financial compliance platform.</li><li>Configured centralized log collection and correlation across multiple servers for real-time threat detection.</li><li>Conducted vulnerability analysis and performance testing to validate IDS/SIEM reliability at scale.</li></ul>                                                                             | 2024 |
| <b>LightWays - Full-Stack IoT Lighting Control System</b> – TFF, Borj Cédria, Tunisia <ul style="list-style-type: none"><li>Developed a full-stack web IoT system for remote monitoring and control of smart lighting systems.</li><li>Built back-end using PHP and MySQL hosted on Raspberry Pi; front-end using HTML5, CSS3, JavaScript, jQuery, AJAX, and UIKit — mobile-responsive and optimized for real-time interaction.</li><li>Implemented device control via IP-based communication with Arduino modules using JSON.</li></ul> | 2020 |

## HACKATHONS

Participated in 11 national and international hackathons across web, mobile, IoT, and AI domains.

- Hacker l'Épargne - Enda Inter-Arabe** (2026): 2nd place · Award for Perseverance and Consistency.
- Municipality Hackathon** (2019): 2nd place.
- NIT Day Challenge** (2019): 2nd place.
- Google Hash Code** (2019): Ranked 25th in Tunisia.

## VOLUNTEERING

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|------------------------------------------------------------------------------------------|------|
| <b>Visual Content Volunteer</b> – Tunisian Red Crescent, Local Committee of Hammam Chatt | 2020 |
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Created promotional posters and awareness materials using Canva to support local humanitarian campaigns.

## INTERESTS

Beekeeping · Reading · Traveling